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ELECTRICIAN JOB SHEET · PROJECT BASED

Water pump on battery backup: Suntree ATS 2P 63 A and a 150 m cable to the pump

Harmony@Huailan Ecolodge, Chiang Mai · Job no. JOB-HHL-PUMP-01 · Issued by Solar ESS Working Group · 2 Nov 2026

How it is wired today

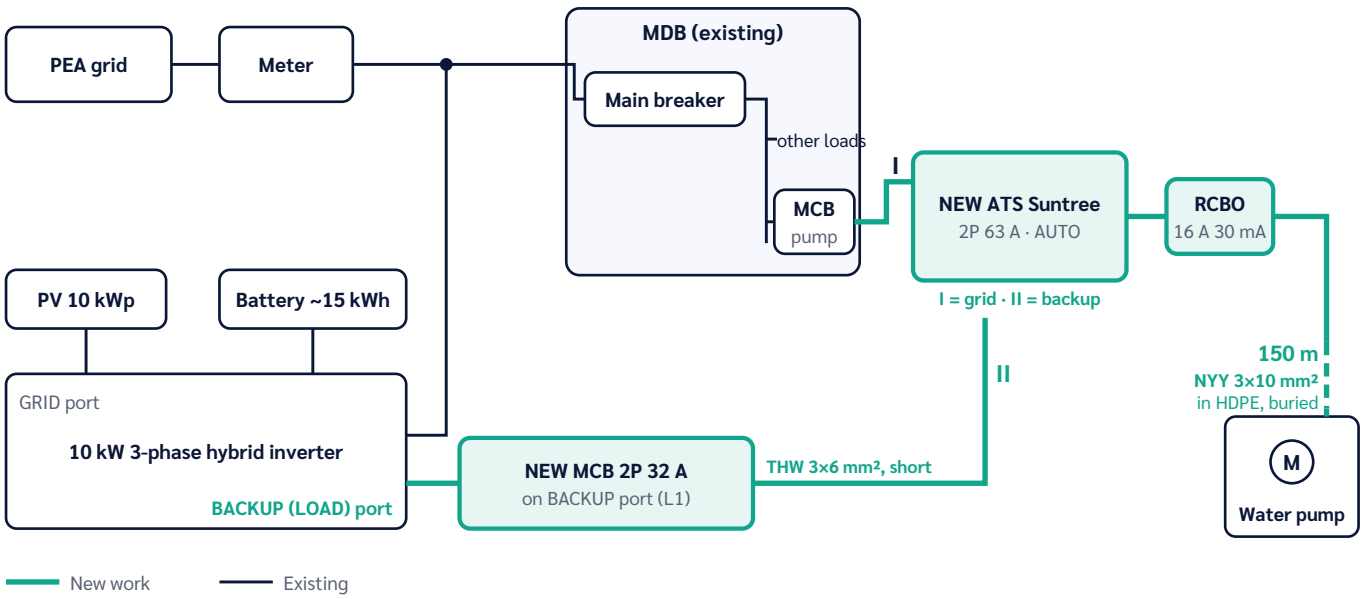
- PEA grid → meter → **inverter GRID port**, and from the same point to the **MDB → all loads**, including the water pump. In a blackout the pump stops.
- The water pump is about **150 m** from the inverter and the MDB.

What we want

- Install a **Suntree 2P 63 A ATS** close to the MDB and inverter, switch set to **AUTO**.
- **Source I (normal):** grid, from the existing pump MCB in the MDB · **Source II (standby):** inverter BACKUP port through a new MCB 2P 32 A.
- Blackout: the ATS transfers to II automatically (3–5 s delay) and the pump runs on the battery · Grid back: returns to I automatically (60 s delay).
- ATS output → RCBO 16 A 30 mA → **NYY 3×10 mm² cable, 150 m**, in buried HDPE pipe → water pump.

Rules that must never be broken: never connect the BACKUP port in parallel with the grid. The ATS must switch both L and N, with a mechanical interlock (break-before-make). Never connect the inverter GRID port to the ATS output.

Single-line diagram



The ATS sits at the supply end (near the MDB and inverter), so both sources are short cables and only one long cable runs from the ATS to the pump. Shown for a single-phase pump up to 1.1 kW.

The 150 m cable to the pump: size and type

Sized for a single-phase 230 V pump up to 1.1 kW (about 7 A running, about 42 A direct-on-line start). Limits: voltage drop ≤ 3% running and about ≤ 15% while starting (important on battery power).

Copper size	Drop running at 7 A	Drop starting at 42 A	Result
6 mm²	7.8 V (3.4%)	47 V (20%)	Fails
10 mm²	4.6 V (2.0%)	28 V (12%)	Passes – use this
16 mm²	2.9 V (1.3%)	17 V (7.6%)	Passes – use if the pump is 1.5–2.2 kW (with soft starter)

- **Cable type:** NYY underground cable, 0.6/1 kV, PVC insulated and sheathed (IEC 60502-1 / TIS), 3 cores 3×10 mm² (L, N, PE), 150 m + 10% spare (≈ 165 m).
- **Do not use THW (60227 IEC 01) for the buried run:** it is not made for wet or underground locations. Use THW only for the short runs indoors.
- **Installation:** in HDPE PE80 pipe Ø 50 mm buried ≥ 0.6 m deep (≥ 0.9 m under driveways), on sand bedding with warning tape above. Pull boxes every ≤ 50 m and at bends; IMC risers where it leaves the ground at both ends.
- **At the pump:** IP65 junction box with cable glands. Earth the pump through the cable PE core; a local earth rod 5/8" × 2.4 m bonded at the pump is recommended.
- **If the existing pump cable is NYY ≥ 10 mm² in good condition** (insulation ≥ 1 MΩ), it may be reused: tell Solar ESS first.

Protection

- **MCB 2P 32 A** on the BACKUP port (the inverter maker's rating) protects the short 6 mm² cable to the ATS.
- **RCBO 1P+N 16 A 30 mA Type A** at the ATS output protects the 10 mm² cable and the pump (pump ≤ 2.2 kW). The pump must keep its own motor overload protection.
- Loop resistance of 150 m × 10 mm² ≈ 0.66 Ω: on the grid the fault current is about 300 A, so the 16 A curve C RCBO trips instantly. On battery the inverter limits short-circuit current itself, and the 30 mA RCD clears earth faults.
- Check that the inverter bonds N–PE in off-grid mode, so the RCBO also works on battery power.

Scope of work

1. **Survey:** pump nameplate (kW, V, current, starting method, pressure switch), existing pump MCB and cable, the 150 m cable route (driveways, rock, roots) and the ATS position. Send to Solar ESS before buying materials.
2. **Isolate (LOTO):** grid main, inverter GRID/BACKUP AC breakers, inverter, battery and PV DC isolators. Prove dead.
3. **ATS:** install the Suntree 2P 63 A ATS on DIN rail in an IP54 enclosure near the MDB/inverter. Terminals I (N, L1) = grid (priority), II (N, L1) = backup. Switch to AUTO; set delays where the model allows (transfer about 3–5 s, return about 60 s).
4. **Source I:** move the outgoing cable of the existing pump MCB in the MDB to ATS I (THW 3×6 mm²).
5. **Source II:** new MCB 2P 32 A on the BACKUP port, phase L1; THW 3×6 mm² in ¾" conduit to ATS II.
6. **Pump cable:** dig the trench, lay HDPE Ø 50 mm ≥ 0.6 m deep, pull NYY 3×10 mm² for 150 m; pull boxes every ≤ 50 m; IMC risers; backfill with warning tape.
7. **RCBO** 1P+N 16 A 30 mA at the ATS output; IP65 junction box at the pump; earth conductor and earth rod at the pump.
8. **Neutral:** the BACKUP-port neutral passes through the ATS only; never join it to the MDB neutral bar.
9. **Labels:** at the ATS and MDB, "Water pump on battery backup – live during a grid outage", and mark the buried cable route.
10. **Test and hand over** per the test table, with photos, an as-built diagram and the Solar ESS report form.

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Danger: the inverter BACKUP port stays live from the battery and the solar panels even when the grid is off, and the ATS transfers to backup by itself in a blackout. Switch off the inverter, the battery and the DC isolators and lock out (LOTO) before working on the pump circuit. Locate existing water pipes and buried cables before digging.

Materials (minimum specification)

Premium materials only. Give the brand and size of every item in your quote.

#	Item	Minimum specification	Qty
1	ATS	Suntree ATS 2P 63 A 230 VAC, DIN-rail, AUTO/MANU switch, I / II / Load ON indicators (or equal); break-before-make with built-in interlock, IEC 60947-6-1	1
2	ATS enclosure	Steel IP54 with DIN rail, terminals and space for the RCBO	1
3	MCB on BACKUP port	2P 32 A curve C 6 kA	1
4	RCBO	1P+N 16 A 30 mA Type A, 6 kA	1
5	Short indoor cables	THW 6 mm² (black, blue, green/yellow) in ¾" IMC/EMT	As measured
6	Pump cable	NYY 0.6/1 kV 3×10 mm² copper (16 mm² if the pump is 1.5–2.2 kW)	≈ 165 m
7	Underground pipe	HDPE PE80 Ø 50 mm, sand bedding, warning tape	≈ 150 m
8	Pull boxes / risers	Concrete or plastic pull boxes every ≤ 50 m and at bends; 1½" IMC risers	As measured
9	At the pump	IP65 junction box, cable glands, waterproof flexible conduit, earth rod 5/8" × 2.4 m with earth wire	1 set
10	Consumables	Lugs, ferrules, cable markers, warning labels, cable ties, anchors	1 set

Tests (record every reading)

#	Test	Pass when
1	Insulation of the 150 m cable at 500 VDC (L–N, L–PE, N–PE), before and after backfill	≥ 1 MΩ
2	Resistance of each core and PE continuity to the pump body	≈ 0.3 Ω per core
3	Earth rod resistance at the pump	≤ 5 Ω
4	Grid normal: ATS on I, pump running; record voltage at the pump running and while starting	≥ 207 V running, ≥ 195 V starting
5	Simulated blackout: open the grid main; ATS transfers to II; pump runs on battery 10 min, restart 3 times; record starting current and voltage at the pump	No trip, ≥ 195 V
6	Grid back: ATS returns to I after the delay	Automatic return
7	RCBO: test button and RCD tester, on grid and on battery	Trips ≤ 300 ms

To confirm on site

- Pump: kW, 1- or 3-phase, current, starting method. Above 1.1 kW or a starting current above about 40 A: a soft starter and 16 mm² cable are needed; tell Solar ESS.
- If the pump is three-phase: 4P ATS, 4P MCB and 5-core NYY; tell Solar ESS first.
- The 150 m route: driveway crossings, soil or rock, any existing pipe that can be reused.
- Existing pump MCB and cable; the inverter's off-grid N–PE setting.

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How to quote

- **Quote two prices:** 1) Labour cost only; 2) Labour & materials (list every material by brand and size).
- Price **trenching and backfill** separately per metre (normal soil / hard soil or rock).
- Pay guide for labour: **฿8,000** for 2–3 people, about 3 days including trench, cable pull, ATS and tests; materials and travel not included.
- Attach your electrician certificates and photos of similar work. Do not change any inverter settings; Solar ESS sets the inverter remotely.
- Send your quote with the form on www.solaress.org/job.html.

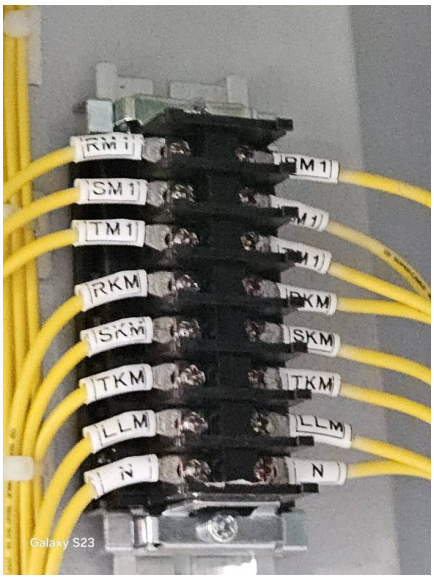
Site photos



Existing MDB (the ATS goes near here)



Inverter room with AC and PV boxes



Existing terminal block (identify before work)

Qualifications and certificates of the Solar ESS team are furnished upon request or upon completion of contract.

Solar ESS Working Group